

COSMIC RAYS FROM SUPER-NOVAE

BAADE & ZWICKY (1934)

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JC - 92 years later

Available at carmeloevoli.github.io

THE PAPERS ON TRIAL TODAY

(I) **On Super-Novae**, Baade and Zwicky, PNAS **20**, 254

invents the class, and weighs one event.

(II) **Cosmic Rays from Super-Novae**, Baade and Zwicky, PNAS **20**, 259

turns that energy into a flux at the Earth.

(III) **Remarks on Super-Novae and Cosmic Rays**, Baade and Zwicky, Phys. Rev. **46**, 76

the fine print, including the part that breaks their own test.

- Papers (I) and (II) were *communicated on the same day*, 19 March 1934, and printed back to back.
- Paper (III) three months later, in June 1934.
- **Ten pages** for **three new ideas**: supernovae, cosmic-ray origins and neutron stars.

THE TWO NAMES OF OUR STORY



Walter Baade (1893–1960)

Mount Wilson Observatory

- German, at Mount Wilson from 1931. The patient observer of the pair.
- In the war, an **enemy alien** barred from war work — so he had the 100-inch to himself, over a blacked-out Los Angeles.
- 1943: resolves the core of M31 → **stellar populations I and II**.
- 1952: the Cepheid mix-up → **the distance scale doubles**.



Fritz Zwicky (1898–1974)

California Institute of Technology

- Swiss, at Caltech from 1925. The idea generator, at volume.
- 1933: **dunkle Materie** in the Coma cluster.
- Never accepted the expanding universe: **tired light is his**, and it sits in Eq. (2) of paper (II).
- 1936: the Palomar 18-inch Schmidt, and the **first systematic supernova search**.

Three papers in 1934, a Nobel-grade idea in each — and **they could not be left in a room together**.

COLLEAGUES, NOT FRIENDS

Jesse Greenstein arrived at Caltech in 1948 to build the astronomy group, and inherited Zwicky:

*“He was **teasable**, and he was **gruff and violent and everything**.”*

*“One of the worst parts of this situation was that he had collaborated with the great Mount Wilson observer and cosmologist Walter Baade in the discovery of supernovae. He and Baade **split violently**. They were a **dangerous pair to put in the same room**; in fact, Zwicky called Baade a Nazi, which he wasn’t. And **Baade said he was afraid that Zwicky would kill him**.”*

PRUD’HOMME: “Very strong.” GREENSTEIN: “Yes. He was a very interesting person.”

p. 26

PRUD’HOMME: “One isn’t necessarily trained to be a collaborator.”

GREENSTEIN: “No. **He was impossible**. I collaborated with him once, through all the years we were together. **I fought with him perhaps ten times a year**. There were always difficulties.”

p. 26

Nine pages of the most consequential astrophysics of the decade came out of that working relationship.

THE THING EVERYBODY CITES THIS FOR

The standard citation: “Baade & Zwicky (1934) proposed that Galactic cosmic rays are accelerated in supernova remnants.”

Three highlighted words, and the 1934 papers support **none of them**.

The one word in that sentence that is genuinely theirs is **super-nova** — they coined it.

- Their cosmic rays are **extragalactic**. Explicitly, in the first sentence.
- There is **no acceleration mechanism**. Not a weak one — none.
- There is **no Galactic magnetic confinement**. The only magnetic field in the papers is the Earth's.
- The Milky Way appears only as a **future** source, one that has been quiet all along.

The plan for the next 30 minutes: follow their arithmetic line by line,
find out why it worked in 1934, and why it cannot work now.

THE ARGUMENT IN FOUR STEPS

1. **Make the sources sporadic.** A constant signal does not require constant sources, if you have enough of them.
2. **Weigh one event.** From the light curve of S Andromedae, get the *total* energy E_T of a supernova.
3. **Give it all to the particles.** Assume a “considerable part” of E_T comes out as cosmic rays.
4. **Multiply.** Energy per event $\times$ rate $\times$ source density $\times$ propagation length = flux at the Earth.

Steps 1 and 4 are still exactly how we do this today.

Steps 2 and 3 are where all the entertainment is.

ACT I — THE IMPASSE

TWO FACTS, ONE CONCLUSION

“Two important facts support the view that cosmic rays are of extragalactic origin, if, for the moment, we disregard the possibility that the earth may possess a very high and self-renewing electrostatic potential with respect to interstellar space.”

(II), opening sentence

“(1) The intensity of cosmic rays is practically independent of time. This fact indicates that the origin of these rays can be sought neither in the sun nor in any of the objects of our own Milky Way.”

(II), p. 259

Fact 2 is the latitude effect: the intensity dips at the geomagnetic equator, so part of the primaries are charged and come from far enough away for the Earth's field to bend them.

The hidden premise. There is **no reservoir**: particles travel ballistically, so the flux at Earth is an instantaneous image of what the sources are doing, and every Galactic candidate on offer in 1934 is *variable* — the sun above all, then novae.

The modern escape is the one they lacked: the Galaxy is bursty at the source and steady at Earth, because $\sim 10^5$ events superpose over a $\sim 10^7$ yr residence time. But interstellar fields will arrive only in 1949 (**Hall; Hiltner**).

THE IMPASSE, IN THEIR OWN WORDS

“Since none of the objects of our Milky Way seem to produce any cosmic rays, these rays probably are not emitted from any of the extragalactic nebulae either, as the spirals among these nebulae are in most respects similar to our own Milky Way. One arrives therefore at one of two hypotheses, either that cosmic rays originate in intergalactic space or that they are survivors from a time when physical conditions in the universe were entirely different from what they are now (Lemaître).”

(II), pp. 259–260

*“On closer scrutiny both hypotheses prove to be very unsatisfactory. On both views one is forced to assume **entirely fantastic processes** as regards the mode of creation of the rays.”*

(II), p. 260

In 1934 “nebula” means galaxy. Throughout, read *nebula* = *galaxy*.

THE IDEA

“In the following we make an entirely new proposal, which, we think, removes some of the major difficulties concerning the origin of cosmic rays. ... If, however, the production of cosmic rays is related to some sporadic process, such as the flare-up of a super-nova, the above-mentioned difficulties disappear.”

(II), p. 260

That is the whole conceptual move, and it is a good one:

- “The Milky Way is not making cosmic rays *now*” only implies “galaxies never make cosmic rays” *if the emission is steady*.
- Make it bursty, and one quiet galaxy is a sampling accident.
- Rate adopted: *one super-nova per galaxy per $\tau = 10^3$ yr*. Hess discovered cosmic rays in 1912. Over ~ 20 yr of observations the chance of a Galactic event is $20/1000 \simeq 2\%$.

Not seeing one was the prediction, not the problem.

WHY THE SUM IS SMOOTH

“From the estimate of one super-nova per galaxy per thousand years it follows that 10^7 super-novae appear per year in the 10^{10} nebulae which are contained in a sphere of 2×10^9 years radius (critical distance derived from the red shift of nebulae). If cosmic rays come from super-novae their intensity in points far away from any individual super-nova will be essentially independent of time.”

(III), p. 76

Check their bookkeeping with one galaxy per cube of side $l = 10^6$ l.y.:

$$N_{\text{gal}} = \frac{4\pi}{3} \left(\frac{2 \times 10^9}{10^6} \right)^3 \simeq 3.4 \times 10^{10} \Rightarrow 3.4 \times 10^7 \text{ events per year.}$$

Consistent — and the sum is smooth as long as each event is spread over $\sim$ days or more at Earth.

ACT II — HOW MUCH ENERGY IS IN ONE SUPER-NOVA?

TWO KINDS OF NEW STAR

“... there exist two well-defined types of new stars or novae which might be distinguished as *common novae* and *super-novae*. No intermediate objects have so far been observed.”

(I), p. 254

Common novae

$$M \simeq -5.8 \quad (2 \times 10^4 L_{\odot})$$

10–20 per year in the Milky Way

Super-novae

$$M \simeq -13$$

“one per stellar system per several centuries”

“... these novae presented the new feature that *at their maximum brightness they emit nearly as much light as the whole nebula in which they originate*. Since ... the absolute total luminosities of extragalactic systems scatter ... around the mean value $M = -14.7$, there is no doubt that we must attribute to this group of novae an individual maximum brightness of the order of $M = -13$.”

(I), p. 255

Remember the number **–13**. It is about to be quietly replaced.

THE TEMPLATE: S ANDROMEDAE

“(3) The super-nova of 1885 in Andromeda is a fair sample. We therefore base our calculations on the characteristics observed for this super-nova.”

(I), p. 256

The arithmetic they used, with the 1934 distance scale:

$$m = 7.5, \quad m - M = 22.2 \quad \implies \quad M_V = -14.7.$$

The class they had just defined sits at $M \simeq -13$.

The “fair sample” they picked sits at $M = -14.7$: a factor 4.8 brighter in visible light.

Hold that factor. In three slides it becomes $4.8^4 \simeq 5 \times 10^2$.

THE VISIBLE OUTPUT, AND A 25-DAY PROBLEM

$$L_V \simeq 6.3 \times 10^7 L_\odot \simeq 2.38 \times 10^{41} \text{ erg s}^{-1} \quad (\text{SN-3})$$

$$E_V \simeq 10^7 \text{ yr of solar radiation} \simeq 1.19 \times 10^{48} \text{ erg} \quad (\text{SN-4})$$

Divide one by the other and you get the effective duration of the event:

$$t_{\text{eff}} \equiv \frac{E_V}{L_V} = 5.0 \times 10^6 \text{ s} \simeq 58 \text{ days}.$$

*“An integration of the light-curve shows that practically the whole visible radiation is emitted **during the 25 days of maximum brightness** and that the total thus emitted is equivalent to 10^7 years of solar radiation of the present strength.”*

(I), p. 255

You cannot radiate for 25 days at a luminosity no greater than L_V and have $E_V/L_V = 58$ days.

The rest of the paper silently uses E_V/L_V , not the stated 25 days. Mistake worth a factor ~ 2.3 — which, as we are about to see, is a rounding error.

FROM VISIBLE TO TOTAL: THE RAYLEIGH–JEANS TRICK

“We assume that the observed L_V corresponds to the *integrated red end* of a black body radiation of the effective temperature T_e ... The integration over the red tail will be taken from the frequency $\nu = 0$ to the violet end of the visible spectrum.”

(I), p. 256

So the visible band is not the peak — it is the *tail*. Integrate Rayleigh–Jeans up to $\nu_V = 7.5 \times 10^{14} \text{ s}^{-1}$:

$$L_V \simeq S \int_0^{\nu_V} \frac{2\pi k T_e \nu^2}{c^2} d\nu = 1.34 \times 10^8 S T_e \quad (\text{SN-6})$$

(recomputes to 1.36×10^8 ; they were good to 2% with a slide rule.)

Then Stefan–Boltzmann on the same surface:

$$L_T = S \frac{ac}{4} T_e^4 \simeq 1.80 \times 10^{-37} L_V^4 S^{-3} \quad (\text{SN-14})$$

$$S \propto R^2, \text{ therefore } L_T \propto L_V^4 R^{-6}.$$

THE RADIUS THAT NOBODY MEASURED

The whole answer hangs on R , the photospheric radius at maximum. It was never observed — it was argued by analogy with common novae:

*“Reasons can be advanced, however, for supposing that probably $R < 100R_{\odot}$, and that almost certainly $R < 400R_{\odot}$... **we feel safe in assuming** that R_m for a super-nova cannot be greater ... **To be conservative**, we shall carry out our calculations for the two cases.”*

(I), p. 257

“To be conservative” + $L_T \propto R^{-6}$ + a factor 4 in an unmeasured radius

⇒ **a factor $4^6 \simeq 4.1 \times 10^3$ in the answer.**

For calibration on Zwicky's confidence in his own estimates: sure that the Japanese would gas Pasadena, he invented a home-made gauze gas mask, called in the Army Corps of Engineers with their tear-gas truck, and went in to come out in thirty seconds. They found him flat on his back on the floor.

*“But that was like him, and **that was a charm of his. He would do anything, because he was convinced that he could figure out how to do it.**”*

Greenstein, Caltech oral history, 1982, p. 28

THE ENERGY CHAIN, EXECUTED TWICE

R	$S \text{ [cm}^2\text{]}$	$T_e \text{ [K]}$	$L_T \text{ [erg s}^{-1}\text{]}$	$E_T = L_T t_{\text{eff}} \text{ [erg]}$
$100R_{\odot}$	6.15×10^{26}	2.89×10^6	2.46×10^{48}	1.23×10^{55}
$400R_{\odot}$	9.84×10^{27}	1.81×10^5	5.98×10^{44}	2.99×10^{51}

Four orders of magnitude, from a factor of four in a number nobody had measured.

Paper (II) then quietly adopts the middle of the range:

$$E_T \simeq 10^{53} - 10^{54} \text{ erg}, \quad L_T \simeq 10^{47} \text{ erg s}^{-1}.$$

Self-consistent, at least: $5 \times 10^6 \text{ s} \times 10^{47} \text{ erg s}^{-1} = 5 \times 10^{53} \text{ erg}$.

Zwicky picks a number and moves on.

INTERLUDE: WHAT A SUPERNOVA ACTUALLY EMITS

Channel	Modern value	B&Z 1934
Optical / radiated light	$\sim 10^{49}$ erg	1.2×10^{48} erg (E_V , close!)
Kinetic energy of ejecta	$\sim 10^{51}$ erg	—
Neutrinos (core collapse)	$\sim 3 \times 10^{53}$ erg	—
Total	$\sim 3 \times 10^{53}$ erg	$10^{53}\text{--}10^{54}$ erg

Their E_T is **right**, to well within their own error bar.

It is right for a channel that would not be discovered for another **53 years** (SN 1987A),
carried by a particle Pauli had proposed **four years earlier** and nobody had seen.

They got the total energy budget of a core collapse essentially correct
by assuming a hot blackbody with a radius they made up. **Physics is not always fair.**

ACT III — FROM ONE STAR TO THE FLUX AT THE EARTH

ENERGY PER PARTICLE: THE ONE HUNDRED MEV

*“The considerations of the preceding paper suggest that in order to produce the stupendous radiation of a super-nova **each particle of mass m must on the average contribute energy of the order***

$$\bar{U} = 0.1 mc^2 ,$$

*which per proton corresponds to an energy of approximately 10^8 volts. Individual photons or material particles ejected from the super-nova may of course possess energies much greater than $\bar{U}$. We therefore feel justified in advancing **tentatively** the hypothesis that cosmic rays are produced in the super-nova process.”*

(II), p. 260

Where 0.1 comes from: with $M_* = 10^{33} - 10^{35}$ g and $E_T = 10^{53} - 10^{54}$ erg,

$$\bar{U} = \frac{E_T}{M_*/m} = \left(\frac{E_T}{M_* c^2} \right) mc^2 \quad \text{spans} \quad 10^{-3} \lesssim \frac{E_T}{M_* c^2} \lesssim 1 .$$

0.1 is the geometric middle of a three-decade range. It is a budget, not a spectrum.

THE STEP THAT IS NOT THERE

*“It also seems reasonable to assume that **a considerable part of the total radiation E_T is emitted in the form of very hard rays or energetic particles.** During the passage through the extremely tenuous material parts of the super-nova some of the cosmic rays will be stopped or softened ... The escape of the cosmic rays must be pictured as a **large-scale analogue to the escape of the various hard rays from a radioactive substance.**”*

(II), p. 260

In modern notation this entire paragraph is $E_{CR} = \eta_{CR} E_T$ with $\eta_{CR} \sim 1$.

- The particles are **born** energetic. Nothing accelerates anything.
- The only physics in the escape is *absorption* on the way out.
- Diffusive shock acceleration is **43 years away** (1977–78). Fermi’s stochastic mechanism is **15 years away** (1949).

They proposed a power supply, not an accelerator.

THE PROPAGATION LENGTH, COURTESY OF ZWICKY 1933

“It has previously been shown that, if radiation is emitted uniformly throughout the universe at the constant rate of $\mathcal{E}$ erg/cm³ sec., the intensity of this radiation reaching the earth is

$$\sigma = \mathcal{E} D/8, \quad D = 2 \times 10^9 \text{ light years.}$$

This formula was derived on the assumption of a red-shift which is proportional to the distance of the source of emission.”

(II), p. 261

The reference is Zwicky's own *How Far Do Cosmic Rays Travel?* — i.e. **tired light**. $D/8$ is an effective path length before redshift losses eat the energy.

“Although we ourselves are by no means convinced that the universe is expanding, it must be stated that the above results are not in contradiction with the short time scale of the order of 10^9 years demanded by some of the relativistic cosmologies.”

(II), p. 261

THE MASTER EQUATION

Three population assumptions, stated in one sentence each:

1. every event gives $E_{\text{CR}} \simeq E_T = 10^{53} - 10^{54}$ erg to cosmic rays;
2. one event per galaxy per $\tau = 10^3$ yr;
3. galaxies sit one per cube of side $l = 10^6$ l.y., so $n_{\text{gal}} = l^{-3}$.

$$\mathcal{E} = \frac{E_{\text{CR}}}{l^3 \tau} \implies \sigma = \frac{E_{\text{CR}} D}{8 l^3 \tau} \quad (\text{CR-3})$$

Read it the modern way and it is a sentence you have written yourself:

$$\sigma = \underbrace{\frac{E_{\text{CR}}}{\tau}}_{\text{power per galaxy}} \times \underbrace{\frac{1}{l^3}}_{\text{source density}} \times \underbrace{\frac{D}{8}}_{\text{effective path length}}$$

This factorization is the part of the 1934 papers that never went out of date.

THE MOMENT OF TRUTH

Plug in $I^3 = 8.47 \times 10^{71} \text{ cm}^3$, $\tau = 3.156 \times 10^{10} \text{ s}$, $D = 1.892 \times 10^{27} \text{ cm}$:

$$\sigma = 8.85 \times 10^{-57} E_{\text{CR}} = 0.9 \times 10^{-3} - 8.9 \times 10^{-3} \text{ erg cm}^{-2} \text{ s}^{-1}.$$

They print $0.8 \times 10^{-3} - 8 \times 10^{-3}$. The difference is rounding.

Measurement (1933)	$\sigma \text{ [erg cm}^{-2} \text{ s}^{-1}]$
Regener, ion counts	3.53×10^{-3}
Millikan, Bowen & Neher	3.2×10^{-3}
B&Z prediction	$(0.9-8.9) \times 10^{-3}$

“Our hypothesis is therefore in surprisingly good agreement with the direct observations of the intensity of cosmic rays.”

(II), p. 261

HOW SURPRISED SHOULD WE BE?

- The prediction is a **one-decade-wide band**. The data land in the middle of it. A one-decade band placed by hand catches a lot of things.
- That band came from inputs spanning **four decades** (R^{-6}), collapsed to one by choosing $E_T \simeq 10^{53}-10^{54}$ erg.
- The template was the **brightest** object in the class, $4.8\times$ above the class value they had themselves defined, and $L_T \propto L_V^4$ turns that into $\sim 5 \times 10^2$.
- $\eta_{CR} \sim 1$ is **not derived anywhere**.
- Everything cosmological — l , D , the redshift law, the horizon — is 1934 vintage.

The arithmetic is right and the agreement is real. But the allowed range was vastly large, so **a matching choice was practically always available.**

A consistency check, not a test of the hypothesis.

ACT IV — THE PREDICTIONS, AND HOW THEY BROKE ONE

WAIT FOR A GALACTIC ONE

If one goes off at distance r , in units of 10^3 l.y., with $L_T = 10^{47} \text{ erg s}^{-1}$:

$$\Delta\sigma = \frac{L_T}{4\pi(10^3 \text{ l.y.})^2 r^2} \simeq \frac{10^4}{r^2} \text{ erg cm}^{-2} \text{ s}^{-1} \quad \Longrightarrow \quad \Delta\sigma(r = 30) \simeq 11 \text{ erg cm}^{-2} \text{ s}^{-1}.$$

That is $\sim 3000\times$ the entire cosmic-ray background, from one star at the Galactic centre.

“If interest in these questions still prevails at that future time, science will therefore be able to test the correctness of our hypothesis some time during the next thousand years or so, as the occurrence of a super-nova in our own system would multiply the intensity of the cosmic rays by a factor one thousand or more.”

(II), p. 262

92 years in. Interest prevails. Still waiting.

OR DO NOT WAIT: WATCH THE NEIGHBOURS

For a source at $10^6 n$ l.y.: $\Delta\sigma \simeq 0.01/n^2 \text{ erg cm}^{-2} \text{ s}^{-1}$, i.e. a **per-cent-level** effect for $n \sim 1$.

*“... we recommend that observers of cosmic rays be on the lookout for **short-period systematic increases** in the intensity of cosmic rays in order to determine as accurately as possible the time and the direction of the maximum intensity. With such data quickly at hand, **astronomers might be able actually to locate the responsible super-nova** in one of our near-by systems. As there are about one thousand nebulae in the region $0 < n < 10$, one super-nova per year should be expected in this “immediate” neighborhood of ours ... for a period of a few days.”*

(II), p. 262

This is a **multi-messenger transient alert programme**, proposed in 1934, with electroscopes.

They even nominate a candidate event to re-examine: “Regener’s curious balloon observation of March 29, 1933”.

THREE MONTHS LATER, THEY SHOOT IT DOWN THEMSELVES

*“The estimates given are perhaps **too optimistic** in view of the fact that the velocities of different particles are different. ... These time lags would tend to **smear out** the change of intensity caused by the flare-up of individual super-novae. Dr. R. M. Langer in one of our seminars was the first to call attention to the straggling of simultaneously ejected particles.”*

(III), p. 77

For a source at 10^6 l.y., delay relative to light $\Delta t \simeq T/2\gamma^2$:

Particle	Energy	Paper (III)	$T/2\gamma^2$
electron	10^{11} eV	410 s	412 s
electron	10^9 eV	47.6 days	47.7 days
proton	10^{11} eV	44 yr	44.0 yr

A **few-day** optical outburst versus a signal spread over **decades**.

The coincidence test is dead, and the authors killed it, in print, by July 1934.

WHERE THE RATE CAME FROM

The famous “one per galaxy per 10^3 years” rests on this list:

Our own galaxy	1572	(Tycho)
Andromeda	1885	(S And)
Messier 101	1907	

plus six events in the ~500 galaxies of Virgo over thirty years, and six more in isolated nebulae.

*“We wish to emphasize that **all of these finds are chance finds** since a systematic search for super-novae has been organized only recently.”*

(III), p. 76

Modern core-collapse rate: ~ **1–3 per century per galaxy**, so their τ is **10–30× too long**. Paper (I) had itself read “one per stellar system per several centuries” off the same data, then adopted 10^3 yr “**to be conservative**”. Since $\sigma \propto 1/\tau$, that lowered their own prediction by a further ~ 3 .

THE PREDICTIONS THAT LANDED

*“If super-novae are giant analogues to ordinary novae we may expect that ionized gas shells are expelled from them at great speeds. If this assumption is correct, **part of the cosmic rays should consist of protons and heavier ions**. Direct tests by cloud chamber experiments at high altitudes are desirable in order to test this conclusion.”*

(III), p. 77

Correct. Primaries are ~ 90% protons, ~ 9% helium, the rest heavier nuclei — confirmed in the 1940s. And it is offered as an afterthought in a Letter.

*“**With all reserve we advance the view that a super-nova represents the transition of an ordinary star into a neutron star**, consisting mainly of neutrons. Such a star may possess a very small radius and an extremely high density.”*

(II), p. 263

The neutron was two years old. Pulsars are 33 years away. “**With all reserve**” is the single greatest understatement in twentieth-century astrophysics.

HONEST TO THE END

They list their own problems, unprompted:

*“(1) So far we cannot offer any satisfactory explanation of the east-west effect.
(2) It remains to be explained why the dust and gas clouds which lie along the principal plane of our own galaxy do not appreciably absorb the cosmic rays. This point, however, needs further observational testing.”*

(II), p. 263

Point (2) is a serious objection to an *isotropic extragalactic* flux: every primary has to cross the disc to reach us.

And they separate their two claims explicitly:

“Our two conclusions [mass annihilated in bulk; super-novae emit cosmic rays] are essentially independent of each other and should perhaps be judged separately, each on its respective merits.”

(II), p. 263, closing sentence

A referee's dream. Nobody has judged them separately since.

DIFFICULTY (1) WAS NOT A DIFFICULTY

*“From the fact that in the cloud-chamber experiments **no protons or charged particles heavier than electrons** have been observed in any considerable number, one might conclude that the corpuscular component of cosmic rays consists of positive or negative electrons, or both. The characteristics of the **east-west effect** indicate that the **positively charged particles far outnumber the negatives**. However, whether or not these particles are electrons cannot as yet be said with certainty, **since the electrons which are observed in cloud chambers may all be secondary particles formed in the earth’s atmosphere by different primaries.**”*

(II), p. 259

positive (east-west) + electron-mass (cloud chambers) = **primary positrons, in bulk**
— and no exploding ball of ordinary matter can supply those.

Anderson had found the positron in 1932 **one corridor away at Caltech**, working for Millikan — whose 3.2×10^{-3} is one of the two fluxes they match.

- Their own $\bar{U} = 0.1 mc^2$ scales with mass: **protons get ~ 100 MeV, electrons ~ 50 keV**. The energetic component must be heavy.
- Paper (III), three months later: “part of the cosmic rays should consist of **protons and heavier ions**”.

The primaries are positive because they are **protons**.
The one unexplained difficulty on the list was a confirmation.

ACT V — 1934 VERSUS TODAY

RUN THEIR EQUATION WITH OUR NUMBERS

Same formula, $\sigma = P_{\text{CR}} \times n_{\text{gal}} \times \ell_{\text{eff}}$, still extragalactic in both rows.

	$P_{\text{CR}} [\text{erg s}^{-1}]$	$n_{\text{gal}} [\text{cm}^{-3}]$	$\ell_{\text{eff}} [\text{cm}]$	$\sigma_{\text{pred}} [\text{erg cm}^{-2} \text{s}^{-1}]$	$\sigma_{\text{obs}} [\text{erg cm}^{-2} \text{s}^{-1}]$
Baade & Zwicky 1934	$3 \times 10^{42} - 3 \times 10^{43}$	1.2×10^{-72}	2.4×10^{26}	$(0.9-8.9) \times 10^{-3}$	$(3.2-3.5) \times 10^{-3}$
Modern values	3×10^{40}	3.4×10^{-76}	3.3×10^{27}	4×10^{-8}	1.2×10^{-2}

Modern row: $E_{\text{CR}} \simeq 10^{50}$ erg per event (10% of 10^{51} erg), one per century per galaxy, $n_{\text{gal}} \simeq 10^{-2} \text{ Mpc}^{-3}$, $\ell_{\text{eff}} \simeq c/4H_0$; observed $\sigma = c u_{\text{CR}}/4$ with $u_{\text{CR}} \simeq 1 \text{ eV cm}^{-3}$.

The extragalactic version of the hypothesis now misses by a factor $\sim 3 \times 10^5$.

ANATOMY OF A LUCKY AGREEMENT

Every factor moved, and two of them moved a long way in the same direction:

- **Power per galaxy: 10^2 – $10^3 \times$ too high.** E_T was overestimated *and* all of it went to cosmic rays ($\eta_{\text{CR}} \sim 1$ instead of ~ 0.1 , on 10^{51} erg instead of 10^{53}).
- **Source density: $\sim 3 \times 10^3$ too high.** $l = 10^6$ l.y. means ~ 35 galaxies per Mpc^3 against $\sim 10^{-2}$ today. Density goes as (distance scale) $^{-3}$, and Hubble's 1934 calibration put galaxies $\sim 7 \times$ too close — that alone is ~ 350 . The residual ~ 10 is what you are willing to call a galaxy.
- **Path length: $\sim 14 \times$ too short,** the only factor that helps: Zwicky's horizon is $\sim 7 \times$ smaller than c/H_0 , and his $D/8$ is half the fraction of it that $c/4H_0$ is.

That factor of 7 in the distance scale was taken apart in stages.

And who took the **first factor of two** out of it, in 1952, starting the demolition of the source density in his own 1934 paper?

Walter Baade.

S ANDROMEDAE, THEN AND NOW

	Baade & Zwicky 1934	Today
Peak apparent magnitude	$m = 7.5$	$m \simeq 6$
Distance modulus of M31	22.2	$\simeq 24.4$
Peak absolute magnitude	$M_V = -14.7$	$M_V \simeq -18.5$
Type	“giant analogue of a nova”	subluminous Type Ia (SN 1885A)
End state	neutron star	nothing — it is a thermonuclear event

The single object on which the whole energy budget rests
is the **one type of supernova that does not leave a neutron star**,
and is not the population that powers Galactic cosmic rays.

In their defence: the Ia / core-collapse split is 1941 (Minkowski), and they said so themselves —
“only a few underexposed spectra of super-novae are available, and it has not thus far been possible to interpret them”.

WHAT IS NOT IN THE PAPERS

- 👎 **No acceleration mechanism.** $\bar{U} \simeq 0.1 mc^2$ is an energy *budget* per particle. There is no spectrum, no shock, no scattering, no dynamics. Fermi: 1949. Diffusive shock acceleration: 1977–78.
- 👎 **No Galactic magnetic confinement.** The only field in the whole argument is the Earth's, and it is only there to explain the latitude effect. Interstellar fields arrive in 1949 with polarized starlight (Hall; Hiltner; Davis & Greenstein 1951).
- 👎 **No steady Galactic population.** A Milky Way supernova is a *future flash* that would swamp the background 1000× for a few days — not one of 10^4 past events whose particles are still bouncing around the disc.

The modern paradigm needs all three.

The 1934 papers contain none of them, and were never claiming to.

WHAT IT TOOK, PART ONE: 1934–1954

1934	Baade & Zwicky	super-novae, neutron stars, extragalactic cosmic rays
1935	Compton & Getting , PR 47 , 817	galactic rotation must imprint an anisotropy on an extragalactic flux
1941	Minkowski	Type I and Type II separated spectroscopically
1944	Reber , ApJ 100 , 279	the Galaxy mapped in radio — the emission is extended
1949	Fermi , PR 75 , 1169	stochastic acceleration — assumes magnetic retention, derives no halo
1949	Hall; Hiltner	polarized starlight — the Galactic field exists
1949	ter Haar , Science 110 , 285	“only our own galaxy” — the budget in modern form
1950	Kiepenheuer , PR 79 , 738	the radio glow is synchrotron from cosmic-ray electrons
1950	Bradt & Peters , PR 80 , 943	Li, Be, B absent ⇒ evidence against Galactic origin
1951	Cocconi , PR 83 , 1193	the first diffusion model — escape from a finite Galaxy
1951	Davis & Greenstein	grain alignment confirms the field
1951	Westerhout & Oort , BAN 11 , 323	bright diffuse emission at high latitude — no disc does that
1952	Baade	the extragalactic distance scale doubles
1952	Hayakawa , PTP 8 , 571	the disc should glow in π^0 γ-rays — cosmic rays made visible
1953	Ginzburg , UFN 51 , 343	the extended halo , argued from radio
1954	Kaplon, Noon & Racette , PR 96 , 1408	Li, Be, B are there — the grammage is real

Eighteen months apart, ter Haar needs $\tau \simeq 5 \times 10^7$ yr to store the cosmic rays and Bradt & Peters allow 10^6 . Their grammage was right; the density was not — and the light nuclei are found four years later. Bristol (Dainton, Fowler & Kent 1952) claimed them first, at $L/M \gtrsim 1$; the truth is $\simeq 0.25$. **Both early answers overshoot, in opposite directions.**

WHAT IT TOOK, PART TWO: 1958–2013

1958	Kulikov & Khristiansen, JETP 8 , 441	the knee in the air-shower size spectrum
1959	Syrovatskii , Sov. Astron. 3 , 22	the diffusion–loss formalism
1959	Freier, Ney & Waddington, PR 113 , 921	the light-nuclei excess confirmed
1961	Ginzburg & Syrovatskii , PTPS 20 , 1	the synthesis
1965	Badhwar, Devanathan & Kaplon, JGR 70 , 1005	energy dependence of the L/M ratio
1966–69	Jokipii; Wentzel; Kulsrud & Pearce	the microphysics of the diffusion coefficient
1972	Kraushaar et al., ApJ 177 , 341	OSO-3: the diffuse Galactic γ -ray sky — Hayakawa, 20 years on
1973	Webber et al., Ap&SS 24 , 17	Li, Be, B <i>isotopes</i> and the cosmic-ray age
1977	Garcia-Munoz et al.; Hagen et al.	the ^{10}Be clock $\Rightarrow$ the halo is real
1977–78	four groups, independently	diffusive shock acceleration
1987	SN 1987A	$\sim 3 \times 10^{53}$ erg in neutrinos, measured
2013	Fermi-LAT	the pion bump in IC 443 and W44 — protons, in a remnant

Nearly eighty years between the citation and the thing it is cited for.

WHAT DID SURVIVE

The method. Only the meaning of the last factor changed.

1934 — extragalactic

$$\sigma = \frac{E_{\text{CR}}}{\tau} \times \frac{1}{l^3} \times \frac{D}{8}$$

source volume = the observable Universe

smoothing = a **cosmological path length**

Today — Galactic

$$u_{\text{CR}} = \frac{\eta E_{\text{SN}} \mathcal{R}_{\text{SN}}}{V_{\text{gal}}} \times \tau_{\text{esc}}$$

source volume = the Milky Way

smoothing = a **magnetic residence time**

$$\mathcal{R}_{\text{SN}} \simeq 3/\text{century}, E_{\text{SN}} \simeq 10^{51} \text{ erg} \Rightarrow P_{\text{SN}} \simeq 10^{42} \text{ erg s}^{-1}$$

$$\text{required } L_{\text{CR}} \simeq 10^{41} \text{ erg s}^{-1} \Rightarrow \eta \simeq 10\%$$

Same three factors, same logic, same order-of-magnitude style of proof.

Confine the same power in a galaxy instead of the Universe, and the answer comes back.

SCORECARD

Right

- 👍 Super-novae are a distinct class, not bright novae
- 👍 They release a “stupendous” amount of energy
- 👍 $E_T \simeq 10^{53} - 10^{54}$ erg
(correct total — for neutrinos)
- 👍 Ejecta at a sizeable fraction of c
- 👍 They leave **neutron stars**
- 👍 Cosmic rays include **protons and heavier ions**
- 👍 Supernovae are the **power source** of cosmic rays
- 👍 Rare bursts + many sources = steady signal
- 👍 The **energy-budget method** itself

Wrong

- 👎 Cosmic rays are **extragalactic**
- 👎 Photon output $10^{53} - 10^{54}$ erg (it is $\sim 10^{49}$)
- 👎 Rate $1/10^3$ yr per galaxy (20× too low)
- 👎 Galaxy density l^{-3} (3×10^3 too high)
- 👎 Tired-light propagation length
- 👎 “Mass may be annihilated in bulk”
- 👎 The optical/cosmic-ray coincidence test
(retracted by the authors, 3 months later)
- 👎 $\eta_{CR} \sim 1$
- 👎 Every star ends as a super-nova

Two errors of $\sim 10^2$ and $\sim 3 \times 10^3$, pointing opposite ways, **and a factor ~ 10 to spare in the error bar.**

SO WHAT DO WE CITE THEM FOR?

Rare, extremely energetic super-novae, spread over a very large number of galaxies, could produce a cosmic-ray flux that is both **nearly constant** and **of the observed magnitude**.

- The numerical agreement was **fragile** and produced by compensating errors.
- The physical picture — extragalactic, unaccelerated, unconfined — is **not** the modern one.
- They connected a brand-new class of astronomical event to the outstanding problem in particle physics **with an order-of-magnitude energy budget**, and they said out loud how shaky every input was.
- That test is still the first thing we do to any proposed cosmic-ray source:
does the energy add up, at the observed rate, after propagation?

Cite them for the question, not for the answer.

LAST WORD TO THE AUTHORS

“If interest in these questions still prevails at that future time, science will therefore be able to test the correctness of our hypothesis some time during the next thousand years or so ...”

W. Baade & F. Zwicky, PNAS 20, 262 (1934)

Ninety-two years down. **Nine hundred and eight to go.**

APPENDIX

SPECULATION, SELF-LABELLED

“... the frequency of occurrence of one super-nova per galaxy per 10^3 years suggest[s] that *the super-nova process might occur to every star once in its lifetime*, marking perhaps the cessation of its existence as an ordinary star. *We realize that this suggestion is highly speculative* in view of the possibility that the frequency of occurrence of super-novae may depend on time and in view of our complete ignorance with respect to the evolution of the universe.”

(III), p. 76

“If neutrons are produced on the surface of an ordinary star they will *“rain” down towards the center* if we assume that the light pressure on neutrons is practically zero. This view explains the speed of the star’s transformation into a neutron star. *We are fully aware that our suggestion carries with it grave implications regarding the ordinary views about the constitution of stars.*”

(III), p. 77

Every star ends as a super-nova: 🚫 Neutrons rain inward from the surface: 🚫

There is a *neutron star* at the bottom of it: 👍

COUNTING EVENTS IS NOT THE CRITERION

A superposition of bursts is steady only if the pulses **overlap**. With a rate $\mathcal{R}$ of contributing events, each seen at Earth with duration δt , the number arriving at once is

$$\mathcal{N}_{\text{ovl}} = \mathcal{R} \delta t, \quad \text{and fluctuations are} \quad \frac{\Delta \sigma}{\sigma} \sim \mathcal{N}_{\text{ovl}}^{-1/2}.$$

$\mathcal{R} = 10^7 \text{ yr}^{-1}$ means **one arrival every 3 seconds** — and that number alone settles nothing:

Pulse duration at Earth δt	$\mathcal{N}_{\text{ovl}}$	signal
a sharp 1-second burst	0.3	flickers at 0.3 Hz
the source itself, $t_{\text{eff}} \simeq 58 \text{ d} = 5 \times 10^6 \text{ s}$	2×10^6	smooth to 10^{-3}
straggling, $\Delta t \simeq T/2\gamma^2 \simeq 10^5 \text{ yr}$	10^{12}	smooth to 10^{-6}

Last row: 10^{11} eV protons ($\gamma \simeq 10^2$) from a typical $T \sim 10^9 \text{ yr}$ away, using their own paper (III) formula.

The velocity dispersion that **destroys their observational test** is the same effect that **guarantees their constant background**.
Paper (III) reports it only as bad news.

THE SECOND CASE, AND A FACTOR OF TWO

To their credit, they check the opposite extreme: what if the surface is so big and cool that $L_T \simeq L_V$, and there is no hidden energy at all?

$$R = 5.81 \times 10^{14} \text{ cm} \quad \text{reached in one day} \quad v = 6.72 \times 10^9 \text{ cm s}^{-1} \simeq 0.22 c .$$

“... which, per proton, gives the kinetic energy $E_K = m_p v^2 / 2 = 0.05 m_p c^2$.”

(I), Eq. (19), p. 258

$$\frac{1}{2} (0.22)^2 = 0.024. \quad \text{Mistake \#2: a factor of two, printed.}$$

Their point survives it: *even* the boring, cool, no-hidden-energy option needs ejecta at a fifth of the speed of light. Both branches say: **something enormous happens**. And that conclusion is right.

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